ELITE IGCSE ACADEMY

Private Past Paper Solutions

Jun 2025 P2H Worked Solutions

Jun 2025 | P2H

33 questions ▪ question image followed by worked solution

PREPARED BY

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TEACHER EDITION

Worked answers included

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PAST PAPER QUESTION Q#: 1 Marks: 3

TOPIC: **Statistics Toolkit**

Jun 2025 P2H - Jun 2025 | P2H

1 Rio has six cards.
He writes a number on each card so that

the range of the numbers is 10
the median of the numbers is 7.5
the mode of the numbers is 6

Rio arranges the cards so that the numbers are in order of size.

		6		10	14

Three of the numbers are hidden.

Complete the cards above to show the three numbers that are hidden.

(Total for Question 1 is 3 marks)

PAST PAPER SOLUTION

Q#: 1 Marks: 3

TOPIC: **Statistics Toolkit**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION

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Topic check. Statistics Toolkit. The tag is correct.

Solution

The range is 10, and the greatest card is 14:

$$14 - \text{smallest} = 10$$

So the smallest card is 4.
The median is 7.5. For 6 cards, the median is the mean of the 3rd and 4th cards:

$$\frac{6+x}{2} = 7.5$$

$$x = 9$$

The mode is 6, so the other missing card must be 6.

FINAL ANSWER

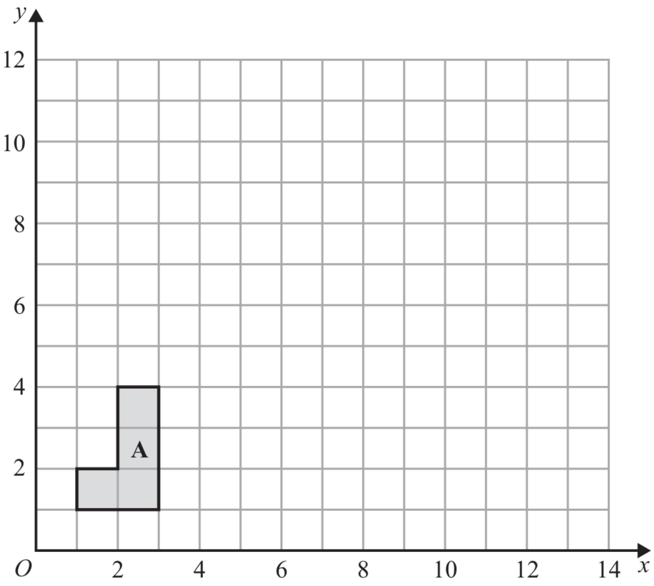
4, 6, 9.

PAST PAPER QUESTION Q#: 2 Marks: 5

TOPIC: **Transformations**

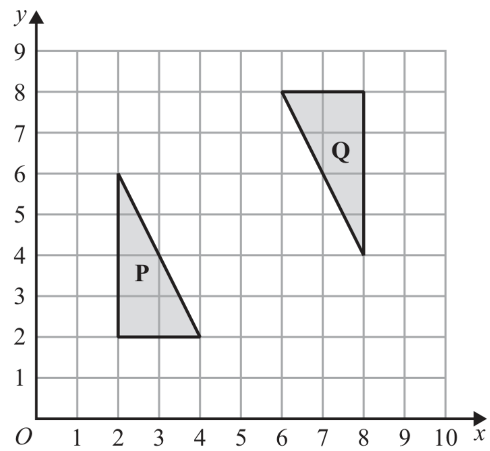
Jun 2025 P2H - Jun 2025 | P2H

2



(a) On the grid, enlarge shape **A** with scale factor 3 and centre (0, 1)

(2)



(b) Describe fully the single transformation that maps triangle **P** onto triangle **Q**

....

....

(3)

(Total for Question 2 is 5 marks)

PAST PAPER SOLUTION

Q#: 2 Marks: 5

TOPIC: **Transformations**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION

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Topic check. Transformations. The tag is correct.

Solution

(a) Enlarge from centre $(0, 1)$ with scale factor 3.
The vertices become

$$(3, 1),\ (9, 1),\ (9, 10),\ (6, 10),\ (6, 4),\ (3, 4)$$

(b) Triangle P is mapped to triangle Q by a $180°$ rotation about $(5, 5)$.

FINAL ANSWER

(a) vertices $(3, 1), (9, 1), (9, 10), (6, 10), (6, 4), (3, 4)$; (b) rotation $180°$ about $(5, 5)$.

PAST PAPER QUESTION

Q#: 3 Marks: 3

TOPIC: **Algebraic Proof**

Jun 2025 P2H - Jun 2025 | P2H

3 Show that $7\frac{1}{3} - 3\frac{4}{7} = 3\frac{16}{21}$

(Total for Question 3 is 3 marks)

PAST PAPER SOLUTION

Q#: 3 Marks: 3

TOPIC: **Algebraic Proof**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION

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Topic check. Algebraic proof. The tag is acceptable because the question asks to show the exact result.

Solution

$$7\frac{1}{3} = \frac{22}{3}, \qquad 5\frac{4}{7} = \frac{39}{7}$$

$$7\frac{1}{3} - 5\frac{4}{7} = \frac{22}{3} - \frac{39}{7}$$

$$= \frac{154}{21} - \frac{117}{21} = \frac{37}{21}$$

$$\frac{37}{21} = 1\frac{16}{21}$$

FINAL ANSWER

$1\frac{16}{21}$.

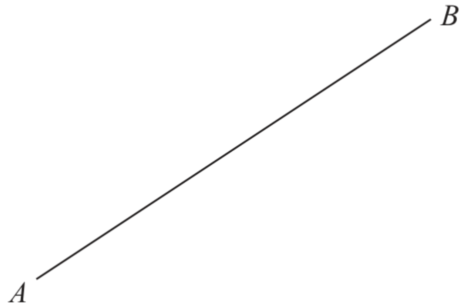
PAST PAPER QUESTION

Q#: 4 Marks: 2

Topic: **Bearings, Scale Drawing & Constructions**

Jun 2025 P2H - Jun 2025 | P2H

4 Using ruler and compasses only, construct the perpendicular bisector of the line *AB*
Show all your construction lines.



(Total for Question 4 is 2 marks)

PAST PAPER SOLUTION Q#: 4 Marks: 2

TOPIC: **Bearings, Scale Drawing & Constructions**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION

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Topic check. Bearings, Scale Drawing & Constructions. The tag is correct.

Construction answer

To construct the perpendicular bisector of AB:

1. Open the compass to more than half the length of AB.
2. Draw equal arcs from A above and below AB.
3. Without changing the compass width, draw equal arcs from B.
4. Join the two arc intersections with a ruler.

FINAL ANSWER

The joined line is the perpendicular bisector of AB.

PAST PAPER QUESTION Q#: 5 Marks: 4

TOPIC: **Congruence, Similarity & Geometrical Proof**

Jun 2025 P2H - Jun 2025 | P2H

5 The diagram shows two similar quadrilaterals, *ABCD* and *EFGH*

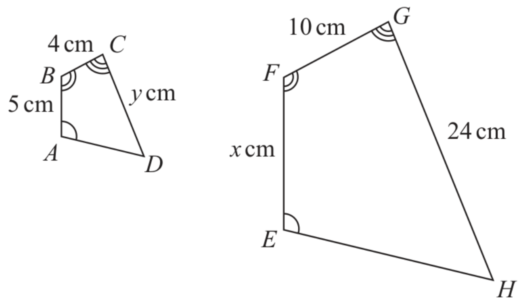

Diagram **NOT** accurately drawn

$AB = 5$ cm $\quad BC = 4$ cm $\quad CD = y$ cm

$EF = x$ cm $\quad FG = 10$ cm $\quad GH = 24$ cm

(a) Work out the value of x

$x =$ ....

(2)

(b) Work out the value of y

$y =$ ....

(2)

(Total for Question 5 is 4 marks)

PAST PAPER SOLUTION

Q#: 5 Marks: 4

TOPIC: **Congruence, Similarity & Geometrical Proof**

Jun 2025 P2H - Jun 2025 | P2H

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Topic check. Congruence, Similarity & Geometrical Proof. The tag is correct.

Solution

The scale factor from $ABCD$ to $EFGH$ is

$$\frac{FG}{BC} = \frac{10}{4} = 2.5$$

Since AB corresponds to EF,

$$x = EF = 5 \times 2.5 = 12.5$$

Since CD corresponds to GH,

$$2.5y = 24$$

$$y = 9.6$$

FINAL ANSWER

$x = 12.5$, $y = 9.6$.

PAST PAPER QUESTION

Q#: 6 Marks: 4

TOPIC: **Ratio Problem Solving**

Jun 2025 P2H - Jun 2025 | P2H

6 Pau, Sam and Tia share £240 in the ratios 3 : 4 : 5

Sam and Tia each give £10 of their share to Pau.

Work out the ratios of the amounts of money that Pau, Sam and Tia now have.
Give your answer in its simplest form.

.... : .... : ....

(Total for Question 6 is 4 marks)

PAST PAPER SOLUTION

Q#: 6 Marks: 4

TOPIC: **Ratio Problem Solving**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION

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Topic check. Ratio Problem Solving. The tag is correct.

Method

The original ratio is

$$\text{Pau} : \text{Sam} : \text{Tia} = 3 : 4 : 5$$

There are

$$3 + 4 + 5 = 12 \text{ parts}$$

Each part is

$$240 \div 12 = 20$$

So the original amounts are

$$60, \ 80, \ 100$$

Sam gives 10 to Pau and Tia gives 10 to Pau, so the new amounts are

$$80, \ 70, \ 90$$

Divide by 10:

$$80 : 70 : 90 = 8 : 7 : 9$$

FINAL ANSWER

The new ratio is $8 : 7 : 9$.

PAST PAPER QUESTION Q#: 7 Marks: 3

TOPIC: **Compound Interest & Depreciation**

Jun 2025 P2H - Jun 2025 | P2H

7 Tim buys a bracelet for 4000 Swiss francs.
The value of the bracelet increases by 7% each year.

Work out the value of the bracelet at the end of 3 years.
Give your answer correct to the nearest Swiss franc.

.... Swiss francs

(Total for Question 7 is 3 marks)

PAST PAPER SOLUTION

Q#: 7 Marks: 3

TOPIC: **Compound Interest & Depreciation**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION

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Topic check. Corrected to Compound Interest and Depreciation. This is repeated percentage growth for 3 years.

Method

The multiplier for a 7% increase is

$$1.07$$

After 3 years,

$$4000(1.07)^3 = 4900.172$$

Correct to the nearest Swiss franc,

$$4900.172 \approx 4900$$

FINAL ANSWER
4900 Swiss francs

PAST PAPER QUESTION

Q#: 8 Marks: 3

TOPIC: **Simultaneous Equations**

Jun 2025 P2H - Jun 2025 | P2H

8 Solve the simultaneous equations

$$3x + 5y = 8$$
$$4x + y = -3.5$$

Show clear algebraic working.

$x =$ ....

$y =$ ....

(Total for Question 8 is 3 marks)

PAST PAPER SOLUTION

Q#: 8 Marks: 3

TOPIC: **Simultaneous Equations**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Simultaneous equations. The tag is correct.

Solution

$$3x + 5y = 8$$

$$4x + y = -3.5$$

From the second equation,

$$y = -3.5 - 4x$$

Substitute into the first equation:

$$3x + 5(-3.5 - 4x) = 8$$

$$3x - 17.5 - 20x = 8$$

$$-17x = 25.5$$

$$x = -1.5$$

Then

$$y = -3.5 - 4(-1.5) = 2.5$$

FINAL ANSWER

$x = -1.5$, $y = 2.5$.

PAST PAPER QUESTION

Q#: 9 Marks: 5

TOPIC: **Graphing Inequalities**

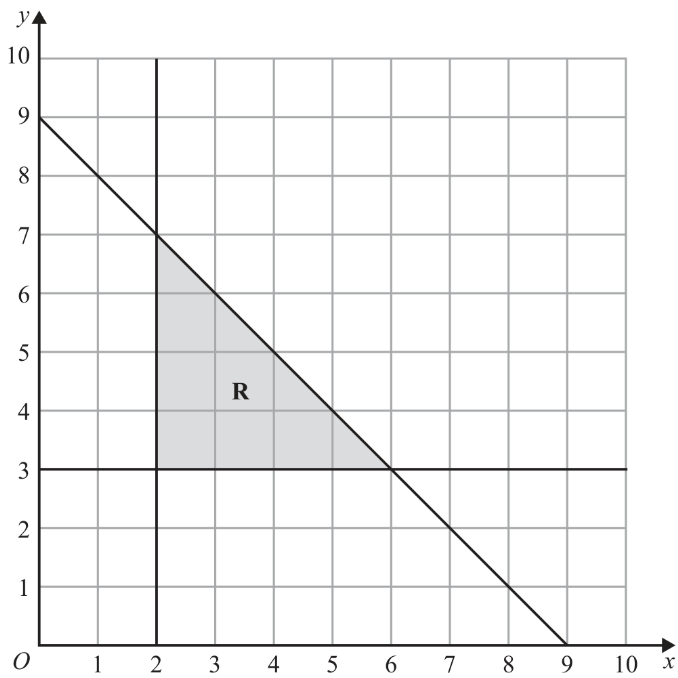
Jun 2025 P2H - Jun 2025 | P2H

9 (a) Solve the inequality $7 - 3t < 2t + 15$

....

(2)

The region **R**, shown shaded in the diagram, is bounded by three straight lines.



(b) Write down three inequalities that define the region **R**

....

....

....

(3)

(Total for Question 9 is 5 marks)

PAST PAPER SOLUTION

Q#: 9 **Marks: 5**

TOPIC: **Graphing Inequalities**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct. This combines solving and identifying inequalities.

Solution

For part (a),

$$7 - 3x < 2x + 15$$

$$-8 < 5x$$

$$x > -\frac{8}{5}$$

For part (b), the boundaries are $x = 2$, $y = 3$, and $x + y = 9$.
The shaded region is

$$x \geq 2$$

$$y \geq 3$$

$$x + y \leq 9$$

FINAL ANSWER

(a) $x > -\frac{8}{5}$. (b) $x \geq 2$, $y \geq 3$, $x + y \leq 9$.

PAST PAPER QUESTION

Q#: 10 Marks: 5

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

Jun 2025 P2H - Jun 2025 | P2H

10 *ABD* and *ABC* are right-angled triangles.

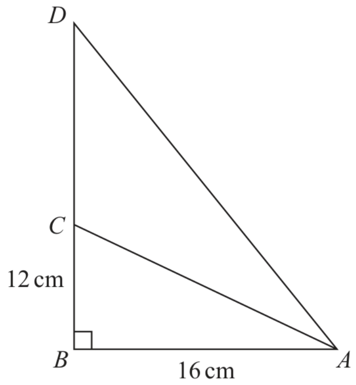


Diagram **NOT** accurately drawn

$AB = 16\,\text{cm}$ $BC = 12\,\text{cm}$ $AD = 1.5 \times AC$

Find the length of *CD*
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 10 is 5 marks)

PAST PAPER SOLUTION

Q#: 10 Marks: 5

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION

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Topic check. Corrected to Right-Angled Triangles - Pythagoras & Trigonometry.

Solution

In triangle ABC,

$$AC^2 = 16^2 + 12^2 = 400$$

$$AC = 20$$

Given $AD = 1.5AC$,

$$AD = 30$$

In triangle ABD,

$$BD^2 = 30^2 - 16^2 = 644$$

$$BD = 25.377\ldots$$

So

$$CD = BD - BC = 25.377\ldots - 12 = 13.377\ldots$$

FINAL ANSWER

13.4 cm.

PAST PAPER QUESTION

Q#: 11 Marks: 3

TOPIC: **Probability Toolkit**

Jun 2025 P2H - Jun 2025 | P2H

11 Khalid has a box of counters.

17 of the counters are red
28 of the counters are blue
the rest of the counters are orange

Khalid is going to take at random a counter from the box.

The probability that Khalid will take an orange counter is $\frac{4}{9}$

Work out the number of orange counters that are in the box.

....

(Total for Question 11 is 3 marks)

PAST PAPER SOLUTION

Q#: 11 Marks: 3

TOPIC: **Probability Toolkit**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Probability Toolkit. The tag is correct.

Solution

There are $17 + 28 = 45$ counters that are not orange.

$$P(\text{orange}) = \frac{4}{9}$$

So

$$P(\text{not orange}) = \frac{5}{9}$$

Let the total number of counters be T.

$$\frac{45}{T} = \frac{5}{9}$$

$$T = 81$$

Number of orange counters:

$$81 - 45 = 36$$

FINAL ANSWER

36.

PAST PAPER QUESTION

Q#: 12 **Marks: 3**

TOPIC: **Expanding Brackets**

Jun 2025 P2H - Jun 2025 | P2H

12 Expand and simplify $3x(2x+5)(7x-4)$

....

(Total for Question 12 is 3 marks)

PAST PAPER SOLUTION

Q#: 12 Marks: 3

TOPIC: **Expanding Brackets**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION

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Topic check. Expanding brackets. The tag is correct.

Solution

$$3x(2x+5)(7x-4) = 3x(14x^2+27x-20)$$

$$= 42x^3+81x^2-60x$$

FINAL ANSWER

$42x^3+81x^2-60x$

PAST PAPER QUESTION

Q#: 13 Marks: 6

TOPIC: **Cumulative Frequency Diagrams**

Jun 2025 P2H - Jun 2025 | P2H

13 The frequency table gives information about the times, in minutes, that 60 people took to complete a puzzle.

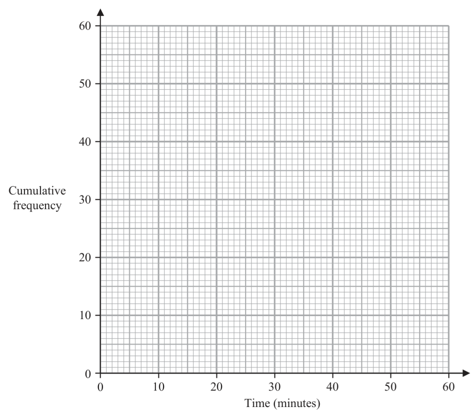
Time (t minutes)	Frequency
$0 < t \leqslant 10$	8
$10 < t \leqslant 20$	13
$20 < t \leqslant 30$	12
$30 < t \leqslant 40$	17
$40 < t \leqslant 50$	7
$50 < t \leqslant 60$	3

(a) Complete the cumulative frequency table.

Time (t minutes)	Cumulative frequency
$0 < t \leqslant 10$	
$0 < t \leqslant 20$	
$0 < t \leqslant 30$	
$0 < t \leqslant 40$	
$0 < t \leqslant 50$	
$0 < t \leqslant 60$	

(1)

(b) On the grid opposite, draw a cumulative frequency graph for your table.



(2)

(c) Use your graph to find an estimate for the median time.

.... minutes

(1)

(d) Use your graph to find an estimate for the number of these people who took more than 45 minutes to complete the puzzle.

....

(2)

(Total for Question 13 is 6 marks)

PAST PAPER SOLUTION

Q#: 13 Marks: 6

TOPIC: **Cumulative Frequency Diagrams**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Cumulative Frequency Diagrams. The tag is correct.

Solution

The cumulative frequencies are

$$8,\ 21,\ 33,\ 50,\ 57,\ 60$$

Plot

$$(0,0),(10,8),(20,21),(30,33),(40,50),(50,57),(60,60)$$

The median is the 30th value.

$$20+\frac{30-21}{33-21}\times 10=27.5$$

At 45 minutes,

$$F(45)\approx 50+\frac{5}{10}(57-50)=53.5$$

So the number taking more than 45 minutes is

$$60-53.5=6.5$$

FINAL ANSWER

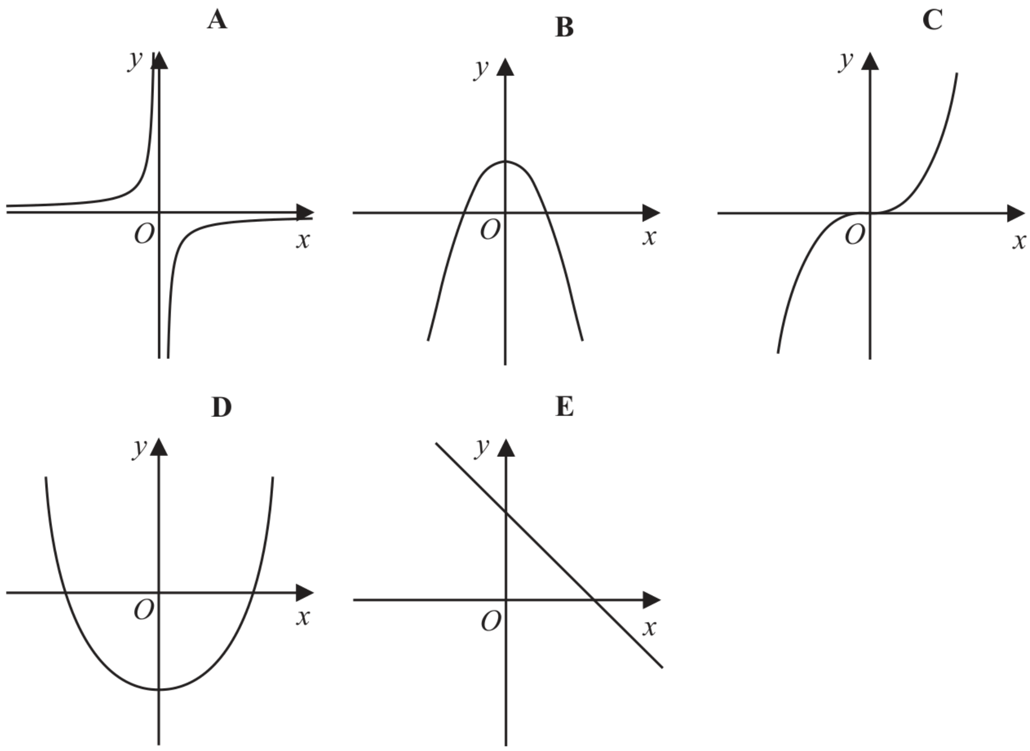
median about 27.5 minutes, and about 7 people.

PAST PAPER QUESTION Q#: 14 Marks: 2

TOPIC: **Graphs of Functions**

Jun 2025 P2H - Jun 2025 | P2H

14 Here are five graphs.



(a) Write down the letter of the graph that could have the equation $y = 5 - x^2$

....

(1)

(b) Write down the letter of the graph that could have the equation $y = 2x^3$

....

(1)

(Total for Question 14 is 2 marks)

PAST PAPER SOLUTION

Q#: 14 Marks: 2

TOPIC: **Graphs of Functions**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected to graphs of functions. The question asks students to match equations to graph shapes.

Solution

(a)

$$y = 5 - x^2$$

This is a quadratic graph with a negative coefficient of x^2, so it is a downward-opening parabola. That is graph **B**.

(b)

$$y = 2x^3$$

This is an increasing cubic graph passing through the origin.
That is graph **C**.

FINAL ANSWER

(a) B, (b) C.

PAST PAPER QUESTION

Q#: 15 Marks: 3

TOPIC: **Algebraic Roots & Indices**

Jun 2025 P2H - Jun 2025 | P2H

15 Simplify fully $\left(\dfrac{2a^3}{10a^7c^2}\right)^{-3}$

....

(Total for Question 15 is 3 marks)

PAST PAPER SOLUTION

Q#: 15 Marks: 3

TOPIC: **Algebraic Roots & Indices**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Algebraic Roots and Indices. The tag is correct.

Method

$$\left(\frac{2a^3}{10a^7c^2}\right)^{-3}$$

First simplify inside the bracket:

$$\frac{2a^3}{10a^7c^2} = \frac{1}{5a^4c^2}$$

A negative power means take the reciprocal:

$$\left(\frac{1}{5a^4c^2}\right)^{-3} = (5a^4c^2)^3$$

$$= 125a^{12}c^6$$

FINAL ANSWER

$125a^{12}c^6$.

PAST PAPER QUESTION

Q#: 16 Marks: 4

TOPIC: **Rearranging Formulas**

Jun 2025 P2H - Jun 2025 | P2H

16 Make t the subject of the formula $c = \dfrac{t^2 + 3}{7 - 8t^2}$

....

(Total for Question 16 is 4 marks)

PAST PAPER SOLUTION

Q#: 16 Marks: 4

TOPIC: **Rearranging Formulas**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Rearranging formulas. The tag is correct.

Solution

$$c = \frac{t^2 + 3}{7 - 8t^2}$$

$$c(7 - 8t^2) = t^2 + 3$$

$$7c - 3 = t^2(8c + 1)$$

$$t^2 = \frac{7c - 3}{8c + 1}$$

FINAL ANSWER

$t = \pm\sqrt{\frac{7c-3}{8c+1}}$

PAST PAPER QUESTION

Q#: 17 Marks: 6

TOPIC: **Differentiation**

Jun 2025 P2H - Jun 2025 | P2H

17 $y = 4x^3 + 5x^2 + 2x$

(a) Find $\dfrac{\mathrm{d}y}{\mathrm{d}x}$

$\dfrac{\mathrm{d}y}{\mathrm{d}x} =$ ....

(2)

(b) Find the coordinates of the turning points on the graph with equation $y = 4x^3 + 5x^2 + 2x$
Show clear algebraic working.

....

(4)

(Total for Question 17 is 6 marks)

PAST PAPER SOLUTION

Q#: 17 Marks: 6

TOPIC: **Differentiation**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION

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Topic check. Corrected to differentiation. The question asks for a derivative and turning points.

Solution

$$y = 4x^3 + 5x^2 + 2x$$

$$\frac{dy}{dx} = 12x^2 + 10x + 2$$

At a turning point,

$$12x^2 + 10x + 2 = 0$$

$$2(6x^2 + 5x + 1) = 0$$

$$(3x + 1)(2x + 1) = 0$$

$$x = -\frac{1}{3} \quad \text{or} \quad x = -\frac{1}{2}$$

When $x = -\frac{1}{3}$,

$$y = -\frac{7}{27}$$

When $x = -\frac{1}{2}$,

$$y = -\frac{1}{4}$$

FINAL ANSWER

$\left(-\frac{1}{2}, -\frac{1}{4}\right)$ and $\left(-\frac{1}{3}, -\frac{7}{27}\right)$.

PAST PAPER QUESTION

Q#: 18 Marks: 2

Topic: **Algebraic Proof**

Jun 2025 P2H - Jun 2025 | P2H

18 Use algebra to show that $0.\dot{7}0\dot{2} = \frac{26}{37}$

(Total for Question 18 is 2 marks)

PAST PAPER SOLUTION

Q#: 18 Marks: 2

TOPIC: **Algebraic Proof**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION

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Topic check. Algebraic proof. The tag is correct.

Solution

Let

$$x = 0.\dot{7}0\dot{2}$$

Then

$$1000x = 702.\dot{7}0\dot{2}$$

Subtract:

$$999x = 702$$

$$x = \frac{702}{999} = \frac{26}{37}$$

FINAL ANSWER

$0.\dot{7}0\dot{2} = \dfrac{26}{37}$.

PAST PAPER QUESTION Q#: 19 Marks: 3

TOPIC: **Solving Inequalities**

Jun 2025 P2H - Jun 2025 | P2H

19 Solve the inequality $4x^2 + 4x - 15 < 0$
Show clear algebraic working.

....

(Total for Question 19 is 3 marks)

PAST PAPER SOLUTION Q#: 19 Marks: 3

TOPIC: **Solving Inequalities**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION

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Topic check. Solving inequalities. The tag is correct.

Solution

$$4x^2 + 4x - 15 < 0$$

Factorise:

$$(2x + 5)(2x - 3) < 0$$

The roots are

$$x = -\frac{5}{2}, \qquad x = \frac{3}{2}$$

The quadratic is negative between the roots.

FINAL ANSWER

$-\frac{5}{2} < x < \frac{3}{2}$.

PAST PAPER QUESTION Q#: 20 Marks: 6

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

Jun 2025 P2H - Jun 2025 | P2H

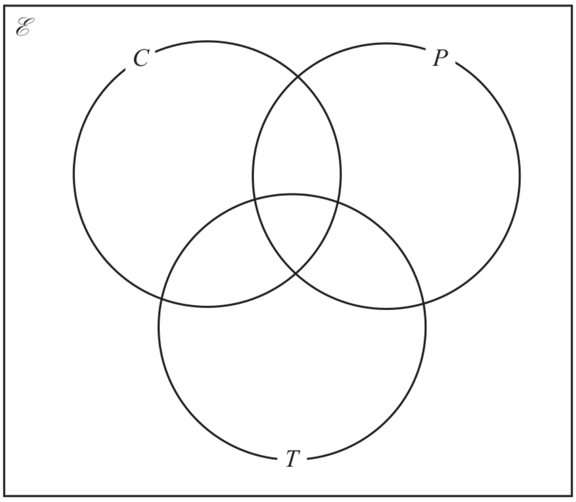
20 120 gardeners were asked if they grow carrots (C) or potatoes (P) or tomatoes (T)

Of these gardeners

43 grow carrots
12 grow carrots and potatoes and tomatoes
18 grow carrots and potatoes
27 grow carrots and tomatoes
32 grow potatoes and tomatoes
29 do not grow carrots or potatoes or tomatoes

The number of these gardeners who grow only potatoes is equal to the number of these gardeners who grow only tomatoes.

(a) Complete the Venn diagram to show this information.



(3)

(b) Find $n(T' \cap C)$

....

(1)

One of the gardeners who grows carrots is chosen at random.

(c) Calculate the probability that this gardener also grows potatoes.

....

(2)

(Total for Question 20 is 6 marks)

PAST PAPER SOLUTION Q#: 20 Marks: 6

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION

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Topic check. Correct.

Method

There are 120 gardeners in total and 29 are outside all three sets, so

$$n(C \cup P \cup T) = 120 - 29 = 91$$

The triple intersection is

$$n(C \cap P \cap T) = 12$$

So the pair-only regions are:

$$C \cap P \text{ only} = 18 - 12 = 6$$

$$C \cap T \text{ only} = 27 - 12 = 15$$

$$P \cap T \text{ only} = 32 - 12 = 20$$

Since 43 grow carrots,

$$C \text{ only} = 43 - 6 - 15 - 12 = 10$$

Let the number who grow only potatoes be y. The number who grow only tomatoes is also y. Now add the regions in the union:

$$10 + y + y + 6 + 15 + 20 + 12 = 91$$

$$63 + 2y = 91$$

$$2y = 28$$

$$y = 14$$

So the Venn regions are:

$$C \text{ only} = 10, \quad P \text{ only} = 14, \quad T \text{ only} = 14$$

$$C \cap P \text{ only} = 6, \quad C \cap T \text{ only} = 15, \quad P \cap T \text{ only} = 20$$

$$C \cap P \cap T = 12, \quad \text{outside} = 29$$

For part (b),

$$n(T' \cap C) = C \text{ only} + C \cap P \text{ only}$$

$$= 10 + 6 = 16$$

For part (c), one gardener who grows carrots is chosen.

$$P(\text{also grows potatoes}) = \frac{n(C \cap P)}{n(C)}$$

$$= \frac{18}{43}$$

FINAL ANSWER
(b) 16, (c) $\frac{18}{43}$

PAST PAPER QUESTION Q#: 20 Marks: 6

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

Jun 2025 P2H - Jun 2025 | P2H

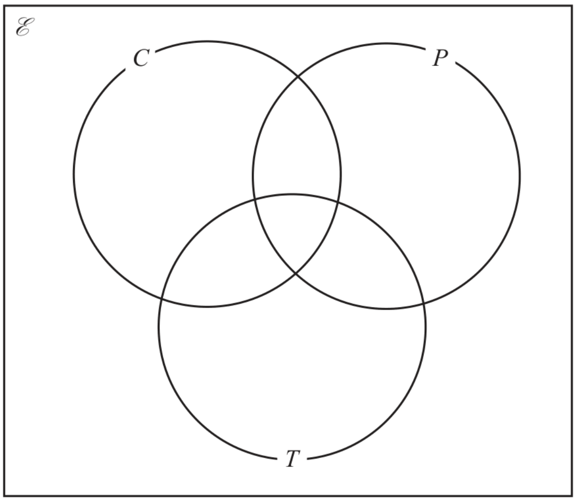
20 120 gardeners were asked if they grow carrots (C) or potatoes (P) or tomatoes (T)

Of these gardeners

43 grow carrots
12 grow carrots and potatoes and tomatoes
18 grow carrots and potatoes
27 grow carrots and tomatoes
32 grow potatoes and tomatoes
29 do not grow carrots or potatoes or tomatoes

The number of these gardeners who grow only potatoes is equal to the number of these gardeners who grow only tomatoes.

(a) Complete the Venn diagram to show this information.



(3)

(b) Find $n(T' \cap C)$

....

(1)

One of the gardeners who grows carrots is chosen at random.

(c) Calculate the probability that this gardener also grows potatoes.

....

(2)

(Total for Question 20 is 6 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 6

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

There are 120 gardeners in total and 29 are outside all three sets, so

$$n(C \cup P \cup T) = 120 - 29 = 91$$

The triple intersection is

$$n(C \cap P \cap T) = 12$$

So the pair-only regions are:

$$C \cap P \text{ only} = 18 - 12 = 6$$

$$C \cap T \text{ only} = 27 - 12 = 15$$

$$P \cap T \text{ only} = 32 - 12 = 20$$

Since 43 grow carrots,

$$C \text{ only} = 43 - 6 - 15 - 12 = 10$$

Let the number who grow only potatoes be y. The number who grow only tomatoes is also y. Now add the regions in the union:

$$10 + y + y + 6 + 15 + 20 + 12 = 91$$

$$63 + 2y = 91$$

$$2y = 28$$

$$y = 14$$

So the Venn regions are:

$$C \text{ only} = 10, \quad P \text{ only} = 14, \quad T \text{ only} = 14$$

$$C \cap P \text{ only} = 6, \quad C \cap T \text{ only} = 15, \quad P \cap T \text{ only} = 20$$

$$C \cap P \cap T = 12, \quad \text{outside} = 29$$

For part (b),

$$n(T' \cap C) = C \text{ only} + C \cap P \text{ only}$$

$$= 10 + 6 = 16$$

For part (c), one gardener who grows carrots is chosen.

$$P(\text{also grows potatoes}) = \frac{n(C \cap P)}{n(C)}$$

$$= \frac{18}{43}$$

FINAL ANSWER

(b) 16, (c) $\frac{18}{43}$

PAST PAPER QUESTION

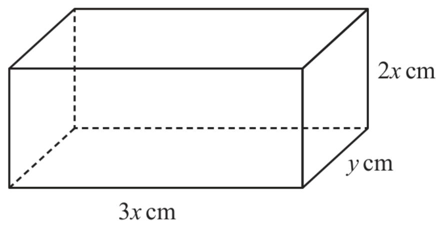
Q#: 21 Marks: 3

TOPIC: **Volume & Surface Area**

Jun 2025 P2H - Jun 2025 | P2H

21 The diagram shows a cuboid.

Diagram **NOT** accurately drawn



The cuboid measures $3x$ cm by $2x$ cm by y cm

The volume of the cuboid is $1014\,\text{cm}^3$
The total surface area of the cuboid is $A\,\text{cm}^2$

Show that $A = 12x^2 + \dfrac{1690}{x}$

You must show all the stages of your working.

(Total for Question 21 is 3 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 3

TOPIC: **Volume & Surface Area**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The cuboid has dimensions

$$3x, \qquad 2x, \qquad y$$

Its volume is 1014 cm^3, so

$$(3x)(2x)(y) = 1014$$

$$6x^2y = 1014$$

$$y = \frac{1014}{6x^2}$$

$$y = \frac{169}{x^2}$$

The total surface area is

$$A = 2(3x \cdot 2x + 3x \cdot y + 2x \cdot y)$$

$$A = 2(6x^2 + 5xy)$$

$$A = 12x^2 + 10xy$$

Substitute $y = \frac{169}{x^2}$:

$$A = 12x^2 + 10x\left(\frac{169}{x^2}\right)$$

$$A = 12x^2 + \frac{1690}{x}$$

FINAL ANSWER

$$\boxed{A = 12x^2 + \frac{1690}{x}}$$

PAST PAPER QUESTION

Q#: 21 Marks: 3

TOPIC: **Volume & Surface Area**

Jun 2025 P2H - Jun 2025 | P2H

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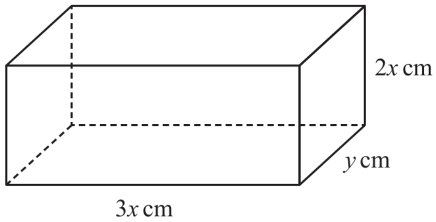
Diagram **NOT** accurately drawn



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Show that $A = 12x^2 + \dfrac{1690}{x}$

You must show all the stages of your working.

(Total for Question 21 is 3 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 3

TOPIC: **Volume & Surface Area**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The cuboid has dimensions

$$3x, \qquad 2x, \qquad y$$

Its volume is 1014 cm^3, so

$$(3x)(2x)(y) = 1014$$

$$6x^2y = 1014$$

$$y = \frac{1014}{6x^2}$$

$$y = \frac{169}{x^2}$$

The total surface area is

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$$A = 2(6x^2 + 5xy)$$

$$A = 12x^2 + 10xy$$

Substitute $y = \frac{169}{x^2}$:

$$A = 12x^2 + 10x\left(\frac{169}{x^2}\right)$$

$$A = 12x^2 + \frac{1690}{x}$$

FINAL ANSWER

$$\boxed{A = 12x^2 + \frac{1690}{x}}$$

PAST PAPER QUESTION

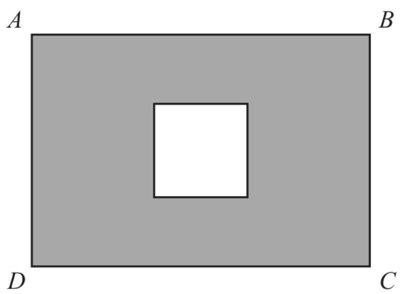
Q#: 22 Marks: 4

TOPIC: **Rounding, Estimation & Bounds**

Jun 2025 P2H - Jun 2025 | P2H

22 The diagram shows a square inside rectangle *ABCD*

Diagram **NOT** accurately drawn



The total area of the region shown shaded in the diagram is X cm²

$AB = 11.5$ cm correct to the nearest 0.5 cm
$BC = 9.2$ cm correct to 2 significant figures
side of square $= 4.1$ cm correct to 2 significant figures

By considering bounds, work out the value of X to a suitable degree of accuracy.
Show your working clearly.

$X =$ ....

(Total for Question 22 is 4 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 4

TOPIC: **Rounding, Estimation & Bounds**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Rounding, Estimation and Bounds. The tag is correct.

Method

The shaded area is

$$X = \text{area of rectangle} - \text{area of square}$$

Bounds:

$$11.25 \le AB < 11.75$$

$$9.15 \le BC < 9.25$$

$$4.05 \le \text{square side} < 4.15$$

Lower bound:

$$11.25(9.15) - 4.15^2 = 85.715$$

Upper bound:

$$11.75(9.25) - 4.05^2 = 92.285$$

So

$$85.715 < X < 92.285$$

Both bounds round to 90 to the nearest 10.

FINAL ANSWER

$X = 90 \text{ cm}^2$ to a suitable degree of accuracy

PAST PAPER QUESTION

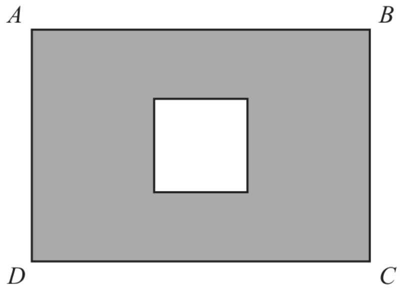
Q#: 22 Marks: 4

TOPIC: **Rounding, Estimation & Bounds**

Jun 2025 P2H - Jun 2025 | P2H

22 The diagram shows a square inside rectangle *ABCD*

Diagram **NOT** accurately drawn



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$AB = 11.5$ cm correct to the nearest 0.5 cm
$BC = 9.2$ cm correct to 2 significant figures
side of square $= 4.1$ cm correct to 2 significant figures

By considering bounds, work out the value of X to a suitable degree of accuracy.
Show your working clearly.

$X =$ ....

(Total for Question 22 is 4 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 4

TOPIC: **Rounding, Estimation & Bounds**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Rounding, Estimation & Bounds is correct.

Method

The shaded area is
$X =$ area of rectangle $-$ area of square
Bounds:
$11.25 \le AB < 11.75$
$9.15 \le BC < 9.25$
$4.05 \le$ side of square < 4.15
Lower bound for X:
$11.25(9.15) - 4.15^2 = 85.715$
Upper bound for X:
$11.75(9.25) - 4.05^2 = 92.285$
So $85.715 \le X < 92.285$. Both bounds round to 90 to the nearest 10.

FINAL ANSWER
$X = 90$ cm^2 to a suitable degree of accuracy.

PAST PAPER QUESTION

Q#: 23 Marks: 4

Topic: **Algebraic Fractions**

Jun 2025 P2H - Jun 2025 | P2H

23 $\dfrac{4x^2-4x-120}{5x^2-180} \div \dfrac{x^2+5x}{10x^2+60x} = p$ where p is an integer.

Find the value of p

Show clear algebraic working.

$p =$ ....

(Total for Question 23 is 4 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 4

TOPIC: **Algebraic Fractions**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected from Ratio Toolkit to Algebraic Fractions.

Method

$$\frac{4x^2 - 4x - 120}{5x^2 - 180} \div \frac{x^2 + 5x}{10x^2 + 60x}$$

Factorise:

$$4x^2 - 4x - 120 = 4(x-6)(x+5)$$

$$5x^2 - 180 = 5(x-6)(x+6)$$

$$x^2 + 5x = x(x+5)$$

$$10x^2 + 60x = 10x(x+6)$$

So

$$\frac{4(x-6)(x+5)}{5(x-6)(x+6)} \div \frac{x(x+5)}{10x(x+6)}$$

$$= \frac{4(x+5)}{5(x+6)} \times \frac{10(x+6)}{x+5} = 8$$

FINAL ANSWER

$p = 8$

PAST PAPER QUESTION

Q#: 23 Marks: 4

TOPIC: **Algebraic Fractions**

Jun 2025 P2H - Jun 2025 | P2H

23 $\dfrac{4x^2-4x-120}{5x^2-180} \div \dfrac{x^2+5x}{10x^2+60x} = p$ where p is an integer.

Find the value of p

Show clear algebraic working.

$p =$ ....

(Total for Question 23 is 4 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 4

TOPIC: **Algebraic Fractions**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected from Ratio Toolkit to Algebraic Fractions.

Method

$$\frac{4x^2-4x-120}{5x^2-180} \div \frac{x^2+5x}{10x^2+60x}$$

Factorise:

$$4x^2-4x-120 = 4(x-6)(x+5)$$

$$5x^2-180 = 5(x-6)(x+6)$$

$$x^2+5x = x(x+5)$$

$$10x^2+60x = 10x(x+6)$$

So

$$\frac{4(x-6)(x+5)}{5(x-6)(x+6)} \div \frac{x(x+5)}{10x(x+6)}$$

$$= \frac{4(x+5)}{5(x+6)} \times \frac{10(x+6)}{x+5} = 8$$

FINAL ANSWER

$p = 8$

PAST PAPER QUESTION

Q#: 24 Marks: 4

Topic: **3D Pythagoras & Trigonometry**

Jun 2025 P2H - Jun 2025 | P2H

24 The diagram shows a square-based pyramid $ABCDE$

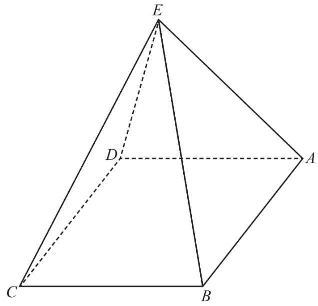

Diagram **NOT** accurately drawn

$EA = EB = EC = ED$

M is the centre of the horizontal square base $ABCD$
Q is the midpoint of AB
Angle $EQM = 80°$

$EA : AB = n : 1$

Find the value of n
Give your answer correct to 3 significant figures.

$n =$ ....

(Total for Question 24 is 4 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 4

TOPIC: **3D Pythagoras & Trigonometry**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Moved from Coordinate Geometry to 3D Pythagoras & Trigonometry.

Method

Let

$$AB = s$$

Since M is the centre of the square base and Q is the midpoint of AB,

$$QM = \frac{s}{2}$$

Since $EA = EB = EC = ED$, the point E is vertically above M.
In right triangle EQM,

$$\tan 80^\circ = \frac{EM}{QM}$$

$$EM = \frac{s}{2}\tan 80^\circ$$

The distance from the centre of a square to a vertex is half the diagonal:

$$AM = \frac{s\sqrt{2}}{2}$$

In right triangle EMA,

$$EA^2 = EM^2 + AM^2$$

$$EA^2 = \left(\frac{s}{2}\tan 80^\circ\right)^2 + \left(\frac{s\sqrt{2}}{2}\right)^2$$

$$EA = s\sqrt{\frac{\tan^2 80^\circ}{4} + \frac{1}{2}}$$

So

$$\frac{EA}{AB} = \sqrt{\frac{\tan^2 80^\circ}{4} + \frac{1}{2}}$$

$$n = 2.92$$

FINAL ANSWER

$n = 2.92$

PAST PAPER QUESTION

Q#: 24 Marks: 4

TOPIC: **3D Pythagoras & Trigonometry**

Jun 2025 P2H - Jun 2025 | P2H

24 The diagram shows a square-based pyramid $ABCDE$

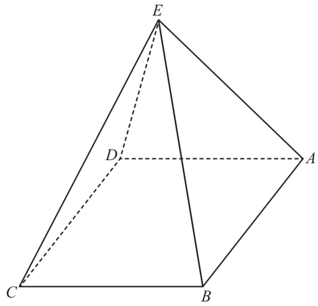

Diagram **NOT** accurately drawn

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Q is the midpoint of AB
Angle $EQM = 80°$

$EA : AB = n : 1$

Find the value of n
Give your answer correct to 3 significant figures.

$n =$ ....

(Total for Question 24 is 4 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 4

TOPIC: **3D Pythagoras & Trigonometry**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Moved from Coordinate Geometry to 3D Pythagoras & Trigonometry.

Method

Let

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In right triangle EMA,

$$EA^2 = EM^2 + AM^2$$

$$EA^2 = \left(\frac{s}{2}\tan 80^\circ\right)^2 + \left(\frac{s\sqrt{2}}{2}\right)^2$$

$$EA = s\sqrt{\frac{\tan^2 80^\circ}{4} + \frac{1}{2}}$$

So

$$\frac{EA}{AB} = \sqrt{\frac{\tan^2 80^\circ}{4} + \frac{1}{2}}$$

$$n = 2.92$$

FINAL ANSWER

$n = 2.92$

PAST PAPER QUESTION

Q#: 25 Marks: 6

TOPIC: **Completing the Square**

Jun 2025 P2H - Jun 2025 | P2H

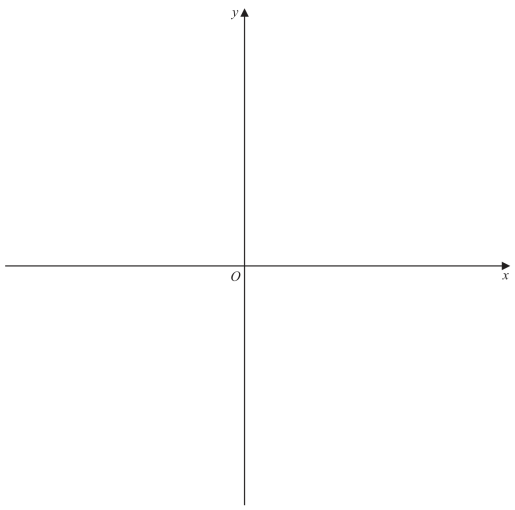
25 (a) Write $28 + 24x - 6x^2$ in the form $a - b(x - c)^2$ where a, b and c are integers.

....

(3)

(b) On the axes below, sketch the graph of $y = 28 + 24x - 6x^2$

Show clearly the coordinates of the turning point and the coordinates of the point of intersection of the graph with the y-axis.



(3)

(Total for Question 25 is 6 marks)

PAST PAPER SOLUTION

Q#: 25 Marks: 6

TOPIC: **Completing the Square**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

(a)

$$28 + 24x - 6x^2 = -6x^2 + 24x + 28$$

$$= -6(x^2 - 4x) + 28$$

$$= -6\left((x-2)^2 - 4\right) + 28$$

$$= 52 - 6(x-2)^2$$

(b)

$$y = 52 - 6(x-2)^2$$

The graph is a downward-opening parabola.
The turning point is

$$(2, 52)$$

The point of intersection with the y-axis is found by putting $x = 0$:

$$y = 28$$

So the y-intercept is

$$(0, 28)$$

FINAL ANSWER

(a) $52 - 6(x-2)^2$. For the sketch, show a downward parabola with turning point $(2, 52)$ and y-intercept $(0, 28)$.

PAST PAPER QUESTION

Q#: 25 **Marks: 6**

TOPIC: **Completing the Square**

Jun 2025 P2H - Jun 2025 | P2H

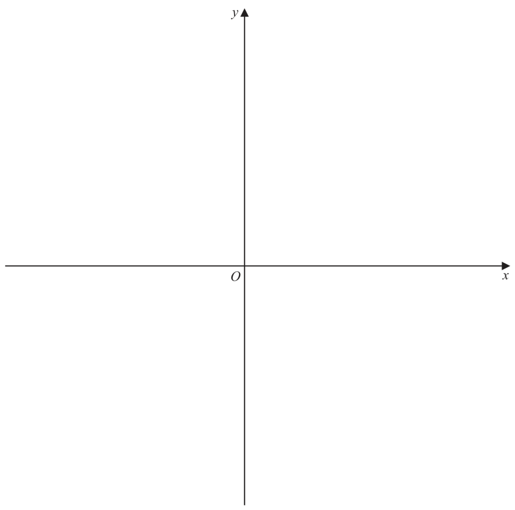
25 (a) Write $28 + 24x - 6x^2$ in the form $a - b(x - c)^2$ where a, b and c are integers.

....

(3)

(b) On the axes below, sketch the graph of $y = 28 + 24x - 6x^2$

Show clearly the coordinates of the turning point and the coordinates of the point of intersection of the graph with the y-axis.



(3)

(Total for Question 25 is 6 marks)

PAST PAPER SOLUTION

Q#: 25 **Marks: 6**

TOPIC: **Completing the Square**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

(a)

$$28 + 24x - 6x^2 = -6x^2 + 24x + 28$$

$$= -6(x^2 - 4x) + 28$$

$$= -6\left((x-2)^2 - 4\right) + 28$$

$$= 52 - 6(x-2)^2$$

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FINAL ANSWER

(a) $52 - 6(x-2)^2$. For the sketch, show a downward parabola with turning point $(2, 52)$ and y-intercept $(0, 28)$.

PAST PAPER QUESTION

Q#: 26 Marks: 4

TOPIC: **Area & Volume of Similar Shapes**

Jun 2025 P2H - Jun 2025 | P2H

26 **A** and **B** are two similar solids.

The height of solid **A** is 31 cm
The height of solid **B** is 18.6 cm

Given that

volume of solid **A** – volume of solid **B** = 735 cm^3

find the volume of solid **A**

.... cm^3

(Total for Question 26 is 4 marks)

PAST PAPER SOLUTION

Q#: 26 Marks: 4

TOPIC: **Area & Volume of Similar Shapes**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The heights of solids A and B are 31 cm and 18.6 cm.

$$31 : 18.6 = 5 : 3$$

So the volumes are in the ratio

$$5^3 : 3^3 = 125 : 27$$

The difference in volume is

$$125 - 27 = 98$$

parts.
This difference is 735 cm^3, so one part is

$$\frac{735}{98} = 7.5$$

Volume of solid A:

$$125 \times 7.5 = 937.5$$

FINAL ANSWER

$\boxed{937.5\ \text{cm}^3}$

PAST PAPER QUESTION

Q#: 26 **Marks: 4**

TOPIC: **Area & Volume of Similar Shapes**

Jun 2025 P2H - Jun 2025 | P2H

26 **A** and **B** are two similar solids.

The height of solid **A** is 31 cm
The height of solid **B** is 18.6 cm

Given that

volume of solid **A** – volume of solid **B** = 735 cm^3

find the volume of solid **A**

.... cm^3

(Total for Question 26 is 4 marks)

PAST PAPER SOLUTION

Q#: 26 Marks: 4

TOPIC: **Area & Volume of Similar Shapes**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The heights of solids A and B are 31 cm and 18.6 cm.

$$31 : 18.6 = 5 : 3$$

So the volumes are in the ratio

$$5^3 : 3^3 = 125 : 27$$

The difference in volume is

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This difference is 735 cm^3, so one part is

$$\frac{735}{98} = 7.5$$

Volume of solid A:

$$125 \times 7.5 = 937.5$$

FINAL ANSWER

$\boxed{937.5 \text{ cm}^3}$